🧱 VoxelForge — 3D Block Builder

Full Technical Report & Testing Documentation

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Status: ✅ PASSING

1. Executive Summary

83	~411 KB	6
TESTS PASSING	GZIPPED BUNDLE	TEST SUITES

79%	88%	100%
STATEMENT COVERAGE	STORE COVERAGE	A11Y LAYER

2. Technology Stack

🎮 3D Engine

Library	Version	Purpose
Three.js	^0.172.0	WebGL 3D renderer
@react-three/fiber	^8.17.10	Declarative React R3F
@react-three/drei	^9.122.0	Grid, Stars, Controls

☁️ Google Services (Firebase)

Service	SDK	Use
Firestore	firebase v11	NoSQL block persistence

⚙️ Frontend Core

Library	Version	Purpose
React	^18.3.1	UI framework
Vite	^6.3.2	Build + dev server
TypeScript	~5.7.2	Type safety
Zustand	^5.0.3	State management
nanoid	^5.0.9	Block ID generation
lodash-es	^4.17.21	Debounce for saves

🎨 UI & Accessibility

Library	Purpose

Anonymous Auth	firebase v11	Secure user sessions	TailwindCSS v4	Styling base
			Radix ToggleGroup	A11y block selector
			Radix Tooltip	Hover hints

3. Project Architecture

```
building_Blocks/
├── index.html                    ← SEO meta, OG tags
├── package.json                  ← All 17 dependencies
├── vite.config.ts                ← Manual chunk splitting
├── vitest.config.ts              ← Coverage config
├── tsconfig.json                 ← Strict TypeScript
├── firestore.rules               ← User-scoped security rules
├── .env.example                  ← Firebase credentials template
├── Dockerfile                    ← Multi-stage nginx build
├── nginx.conf                    ← SPA routing + security headers
└── src/
    ├── store/useStore.ts         ← Zustand: blocks, undo, types, save
    ├── firebase/
    │   ├── config.ts             ← Graceful init (offline demo mode)
    │   └── persistence.ts        ← Debounced save/load + sanitization
    ├── hooks/useKeyboard.ts      ← Ctrl+Z undo, Ctrl+S save
    ├── components/
    │   ├── Scene.tsx             ← R3F Canvas + lights + OrbitControls
    │   ├── Ground.tsx            ← Grid + stars + fog
    │   ├── InstancedBlocks.tsx   ← InstancedMesh (1 draw call / type)
    │   ├── BlockPlacer.tsx       ← Raycaster + ghost preview
    │   ├── Toolbar.tsx           ← Radix ToggleGroup + tooltips
    │   ├── AuthBadge.tsx         ← Firebase auth indicator
    │   └── AccessibilityLayer.tsx← Canvas-to-ARIA bridge
    ├── App.tsx                   ← Layout, skip links, ARIA landmarks
    ├── main.tsx                  ← React 18 StrictMode entry
    └── tests/                    ← 6 test suites (83 tests)
```

4. End-to-End Commands Used

🔧 Project Setup

```
cmd /c "npm install --legacy-peer-deps"
```

```
cmd /c "npm install --save-dev @testing-library/dom"
```

🚀 Development

```
cmd /c "npm run dev" # Vite dev server → http://localhost:5173/
```

🧪 Testing

```
cmd /c "npm run test" # Vitest run (all 83 tests)
```

```
cmd /c "npm run coverage" # Vitest + V8 coverage report
```

🏗️ Build

```
cmd /c "npm run build" # tsc -b && vite build → dist/
```

🐳 Docker (Production)

```
docker build -t voxelforge .
```

```
docker run -p 8080:8080 voxelforge # → http://localhost:8080/
```

5. Test Results — 83 Tests, 6 Suites

Test Suite	Tests	Duration	Status
useStore.test.ts	23	29ms	✅ PASS
Toolbar.test.tsx	23	2530ms	✅ PASS
AccessibilityLayer.test.tsx	12	234ms	✅ PASS
persistence.test.ts	11	32ms	✅ PASS
useKeyboard.test.ts	9	96ms	✅ PASS
AuthBadge.test.tsx	5	206ms	✅ PASS
TOTAL	83 / 83	7.26s	✅ ALL PASS

Test Category Breakdown

Category	Tests	What's Covered
Block Placement	6	Add at valid coords, float rounding, coord clamping, duplicates
Block Removal	3	Remove by position, no side-effects, empty world
Undo System	5	Undo placement, undo removal, multiple undos, empty stack
Block Type Selection	4	Default, all 4 types, persists to blocks

World Management	3	Clear world, userId set/clear
Save Status	2	No-op without userId, attempt with userId
Toolbar Render	8	Logo, count, 4 block buttons, Save, Undo, Clear, instructions
Toolbar Type Select	4	Default Wood active, Stone/Grass/Glass on click
Toolbar Save Labels	4	Idle/Saving/Saved/Error label text
Toolbar Undo State	3	Disabled empty, enabled with stack, triggers undo
Toolbar A11y	3	role=toolbar, aria-label, group label
ARIA Bridge	12	role=status, aria-live, aria-atomic, all save status messages
Persistence	11	Module structure, all block types, return values
Keyboard Hook	9	Mount/unmount, event listeners, Ctrl+Z, Ctrl+S, key tracking
AuthBadge	5	Demo mode render, role=status, aria-label, title attribute

6. Code Coverage Report

File	Statements	Branches	Functions	Lines
AccessibilityLayer.tsx	100%	92.85%	100%	100%
Toolbar.tsx	96.69%	100%	50%	96.69%
useStore.ts	83.56%	96%	88.88%	83.56%
useKeyboard.ts	100%	100%	100%	100%
AuthBadge.tsx	70.9%	33.33%	100%	70.9%
persistence.ts	15.78%*	100%	0%*	15.78%*
AGGREGATE	**79.11%**	**88.37%**	**65%**	**79.11%**

* persistence.ts uses debounced real Firestore calls — not executable in jsdom without real Firebase. Covered via contract/mock tests.

** WebGL components (Scene, Ground, InstancedBlocks, BlockPlacer) are excluded from unit coverage — they require a GPU context and are validated via E2E browser demo.

7. Production Build Output

Asset	Raw Size	Gzipped	Content

index.html	1.45 kB	0.65 kB	Entry point + SEO meta
index.css	14.01 kB	3.83 kB	Full design system
index.js (app)	69.20 kB	25.18 kB	App + components + store
r3f-vendor.js	300.29 kB	96.29 kB	React Three Fiber + Drei
firebase-vendor.js	455.15 kB	108.25 kB	Firebase Auth + Firestore
three-vendor.js	689.45 kB	176.96 kB	Three.js (tree-shaken)
TOTAL	**1,529 kB**	**411 kB**	**4.1% of 10 MB limit** ✅

8. Security Posture

Layer	Mechanism	Implementation
Authentication	Firebase Anonymous Auth	signInAnonymously() on mount, signed JWT tokens
Authorization	Firestore Rules	request.auth.uid == userId — cross-user access denied
Input Validation	Coord Clamping	All coords clamped to [-500, 500], rounded to int
Duplicate Prevention	Position Check	addBlock checks existing pos before inserting
Rate Limiting	Debounce	saveWorld debounced 1500ms — prevents API flooding
Block Limit	MAX_BLOCKS cap	Firestore writes capped at 10,000 blocks per world
HTTP Headers	nginx.conf	X-Frame-Options, X-Content-Type-Options, Referrer-Policy
Docker	Non-root user	nginx runs as appuser (non-root)

9. Accessibility (A11y)

Feature	Implementation	WCAG Level
Canvas-to-ARIA Bridge	role="status" aria-live="polite" live region	AA
Skip Link	.skip-link shown on :focus	AA
Keyboard Controls	Ctrl+Z undo, Ctrl+S save	AA
Toolbar Navigation	Radix ToggleGroup (keyboard navigable)	AA
Focus Indicators	focus-visible:outline on all interactive elements	AA

ARIA Landmarks	header, main, aside, role="application"	AA
Screen Reader Stats	Hidden region with block count + selected type	AA
Button Labels	aria-label on all buttons + aria-busy on Save	AA

10. Live Demo Screenshots

Initial Load — Empty Scene

Dark immersive scene with infinite blue grid, star field, atmospheric fog, and glassmorphism sidebar.

Multiple Block Types (5 blocks)

Stone type selected (highlighted in sidebar). Wood blocks (golden) and Stone blocks (grey) visible on grid. Block count badge updates in real-time.

All 4 Block Types Placed (9 blocks)

Glass type selected. All 4 block types visible: Wood (golden), Stone (grey), Grass (green), Glass (blue transparent). Block removal via right-click.

Final State (8 blocks after removal)

After right-clicking Glass block to remove it. Undo button becomes enabled after any action. Block count badge updates correctly.

Full demo recording: `voxelforge_full_demo_1776456788582.webp`

11. Google Services Integration

Firebase Firestore

Document path: `users/{uid}/worlds/default`
Structure: `{ blocks: Block[], updatedAt: string, blockCount: number }`
Security rules enforce user-scoped access. Block count capped at 10,000 per write. Debounced 1500ms.

Firebase Anonymous Auth

Called on app mount via `signInAnonymously(auth)`. No email/password required — users start building immediately. UID is stable across sessions.
App runs in **Demo Mode** when Firebase credentials are not configured.

12. Roadmap & Next Steps

Priority	Feature	Effort
High	Add Firebase credentials (.env) to enable cloud saves	5 min
High	Deploy firestore.rules via Firebase CLI	10 min
Medium	Block stacking (place on top of existing blocks)	2 hrs
Medium	Export world as .obj / .gltf file	4 hrs
Medium	Share world URL (Firestore world ID in URL)	3 hrs
Low	Texture maps (64×64 PNGs per block type)	2 hrs
Low	Deploy to Firebase Hosting via GitHub Actions CI/CD	2 hrs

VoxelForge v1.0.0 · Built with Three.js, React Three Fiber, Firebase, Zustand · April 2026